

# Anthony Meza

Ph.D. Candidate

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## EDUCATION

### Massachusetts Institute of Technology & Woods Hole Oceanographic Institution

*Ph.D. in Physical Oceanography | Advised by Dr. Geoffrey Gebbie*

*Dissertation: Forced Responses of the Global Ocean Circulation and Implications for Tracer Distributions*

### University of California, Irvine

*B.S. in Mathematics*

Cambridge, MA

*Expected Nov 2026*

Irvine, CA

*2018–2021*

## PUBLICATIONS

- **Meza, A.**, & Gebbie, G. (2025). Wind-driven mid-depth Pacific cooling in a dynamically consistent ocean state estimate. *Journal of Geophysical Research: Oceans*. doi.org/10.1029/2025JC022462
- **Meza, A.**, Drake, H. & Gebbie, G. (*In prep*). Sea ice-driven decline of abyssal ocean ventilation in a high-resolution coupled climate model.
- **Meza, A.**, Bhuyan, P., Zheng, Z. & Gebbie, G. (*In prep*). Passive and dynamic contributions to mean ocean temperature during the Last Glacial Maximum.

## RESEARCH EXPERIENCE

### Woods Hole Oceanographic Institution

*Graduate Research Assistant*

Sep 2021–Present

*Woods Hole, MA*

- Designed physics-based (MITgcm) and Bayesian models to test hypotheses of mechanisms controlling deep ocean circulation and heat content data sparse regions.
- Evaluated high-resolution coupled climate models to quantify the impacts of Antarctic sea ice melt on deep water formation and interior tracer distributions.
- Used ocean reanalysis and satellite products to establish a statistical relationship between remotely-generated near-shore sea surface temperature variability and extreme precipitation events in California.
- Developed Python and Julia tools to process, visualize, and interpret tens of terabytes of climate data in high-performance computing environments.

### Foundation for Resilient Societies

*Technical Consultant*

Jan 2025

*Cambridge, MA*

- Ran and debugged Strategic Energy & Risk Valuation Model (SERVM) simulations to assess adequacy of U.S. electrical grid capacity under varying generation scenarios (e.g., solar adoption).
- Led a team of 12 undergraduate electric grid modeling interns to develop an internal user guide for running SERVM experiments and interpreting model output.

### Los Alamos National Laboratory

*Research Intern*

Jun 2021–Aug 2021

*Los Alamos, NM*

- Implemented and evaluated reduced-precision in the Energy Exascale Earth System Model (E3SM) to reduce computational cost and energy consumption in global climate simulations.

### Institute for Pure and Applied Mathematics & The Aerospace Corporation

*Research Intern*

Jun 2020–Sep 2020

*Los Angeles, CA*

- Designed and implemented reinforcement learning-based methods for adaptive packet routing in satellite network simulations, implemented using PyTorch.

## TECHNICAL PROJECTS

### xndbc

*In development*

*Personal Project*

*github.com/anthony-meza/xndbc*

- A Python package to query, subset and process data from the NOAA National Data Buoy Center (NDBC) and store them as xarray objects.

### Atmospheric Physics using AI

*Apr 2024–Jul 2024*

*Kaggle Competition Participant*

*Remote*

- Ranked 70th percentile on the public leaderboard for emulating subgrid-scale atmospheric processes.
- Developed machine learning architectures with physical constraints in PyTorch and TensorFlow.
- Optimized data pipelines for GPU-accelerated training, significantly increasing model throughput.

## INVITED TALKS

- "How the Ocean Redistributes Heat"

*May 2026*

*Atlantic Oceanographic and Meteorological Laboratory*

## TEACHING EXPERIENCE

- Co-organizer and Instructor  
*High Performance Computing and Data Analysis Workshop*

*October 2024*

- Calculus Instructor  
*MIT-WHOI Joint Program Summer Math Refresher*

*June 2024*

- Partial Differential Equations Instructor  
*MIT-WHOI Joint Program Summer Math Refresher*

*July 2023*

## PROFESSIONAL ACTIVITIES

- Committee Member  
*AMS Committee on Climate Variability and Change*

*2024–Present*

- Graduate Application Mentor  
*Joint Program Applicant Support & Knowledgebase*

*2023–Present*

- Physical Oceanography Representative  
*WHOI Joint Program Student Representative*

*2023–2024*

- Conference Co-Organizer  
*2023 Graduate Climate Conference*

*2023*

- At-Large Representative  
*WHOI Joint Program Student Representative*

*2022–2023*

- Conference Co-Organizer  
*2022 First Generation Summit*

*2022*

- Committee Member  
*UC Irvine Mathematics Inclusive Excellence Committee*

*2020–2021*

## WORKSHOPS AND SUMMER SCHOOLS ATTENDED

- CESM/MOM6 Regional Modeling Workshop  
*NCAR Mesa Laboratory, Boulder, CO*

*May 2025*

- ECCO Summer School  
*Asilomar Conference Center, Monterey, CA*

*May 2025*

- Tracer Mixing in Fluids Across Planetary Scales Summer School  
*Brin Mathematics Research Center, College Park, MD*

*Jul 2024*

## PRESENTATIONS

- **A. Meza**, H. Drake, G. Gebbie. "Projected changes in Southern Ocean surface water mass transformation in an eddy-resolving coupled climate model" Ocean Sciences Meeting, 22–27 February 2026, Glasgow, Scotland. *Poster*.
- **A. Meza**, G. Gebbie. "Wind-driven mid-depth Pacific cooling in a dynamically consistent ocean state estimate" ECCO Summer School, 19–30 May 2025, Asilomar Conference Center, Monterey, CA. *Poster*.
- **A. Meza**, P. Bhuyan, Z. Zheng, G. Gebbie., M. Linz, J. Wenegrat. "Surface to Bottom Connections in Earth's Ocean" Tracer Mixing in Fluids Across Planetary Scales Summer School, 8–19 Jul 2024, Brin Mathematics Research Center, College Park, MD. *Talk*.
- **A. Meza**, H. Seo. "Associations Between Coastally Trapped Waves and Wintertime Precipitation in California" Ocean Sciences Meeting, 18–23 Feb 2024, New Orleans, LA. *Poster*.
- **A. Meza**, H. Seo. "Associations Between Coastally Trapped Waves and Wintertime Precipitation in California" Graduate Climate Conference, 1–3 Nov 2023, Marine Biological Laboratory, Woods Hole, MA. *Poster*.
- **A. Meza**, G. Gebbie. "Drivers of subsurface Pacific cooling in ECCOV4r4" ECCO Annual Meeting 2023, 25 Jan 2023, University of Washington, Seattle, WA. *Virtual Talk*.
- **A. Meza**, G. Gebbie. "Drivers of mid-depth Pacific cooling trends in an ocean reanalysis" AGU Fall Meeting 2022, 2–4 Nov 2023, Chicago, IL. *Poster*.
- **A. Meza**, G. Gebbie. "Drivers of mid-depth Pacific cooling trends in an ocean reanalysis" Graduate Climate Conference, 31 Oct 2022, University of Washington, Seattle, WA. *Poster*.
- C. Tran, **A. Meza**, H.L. Tung, H. Liu. "A Reinforcement Learning Approach to Packet Routing on a Dynamic Network" Joint Mathematics Meeting, 6–9 Jan 2021, Virtual. *Virtual Talk*.

## AWARDS AND HONORS RECEIVED

- |                                                                                                                    |      |
|--------------------------------------------------------------------------------------------------------------------|------|
| • GEM Fellowship<br><i>National Consortium of Graduate Degrees for Minorities in Engineering and Science, MIT</i>  | 2021 |
| • Rose Hills Science & Engineering Scholarship<br><i>Rose Hills Foundation, UC Irvine</i>                          | 2020 |
| • Rose Hills Science & Engineering Scholarship<br><i>Rose Hills Foundation, UC Irvine</i>                          | 2019 |
| • Bellettini School of Physical Science Scholarship<br><i>Maria Rebecca and Maureen Bellettini Fund, UC Irvine</i> | 2019 |
| • Southern California Edison STEM Scholarship<br><i>Southern California Edison, UC Irvine</i>                      | 2019 |

## SKILLS

**Languages:** *Programming:* Python, Julia, Fortran, MATLAB; *Human:* English, Spanish

**Scientific Computing:** NumPy, SciPy, xarray, Pandas, xgcm, Optimization.jl, JuMP.jl, scikit-learn, PyTorch, TensorFlow, keras

**HPC & Dev Tools:** Unix/Linux, OpenMPI, HPC job schedulers (e.g., Slurm), Dask, Git, GitHub

## PROFESSIONAL REFERENCES

Geoffrey Gebbie (ggebbie@whoi.edu)

Henri Drake (hdrake@uci.edu)

Christopher Piecuch (cpiecuch@whoi.edu)